

Advanced Machine Learning

DATA 642

Lecture 15 — Notes and challenges about GANs

*Figures and notes are from the book Mathematics for Machine Learning by A. Aldo Faisal, Cheng Soon Ong, and Marc Peter Deisenroth, Machine Learning A Bayesian and Optimization Perspective, Sergios Theodoridis, Elsevier, and Hands-On Machine Learning with Scikit-Learn Keras and TensorFlow, Aurelien Geron, O'Reilly

The Difficulties of Training GANs

Training GANs poses several challenges that stem from the dynamic interplay between the generator and the discriminator.

- As the training progresses, the generator and discriminator engage in a zero-sum game, each striving to outsmart the other. This adversarial nature can lead to a **Nash equilibrium**, a state where neither player can improve their strategy without the other changing theirs.
- Nash equilibrium is when no player would be better off changing their own strategy, assuming the other players do not change theirs. For example, a Nash equilibrium is reached when everyone drives on the left side of the road: no driver would be better off being the only one to switch sides. Of course, there is a second possible Nash equilibrium: when everyone drives on the right side of the road. Different initial states and dynamics may lead to one equilibrium or the other.
- In this example, there is a single optimal strategy once an equilibrium is reached (i.e., driving on the same side as everyone else), but a Nash equilibrium can involve multiple competing strategies (e.g., a predator chases its prey, the prey tries to escape, and neither would be better off changing their strategy).

Mode collapse

- Authors of the GAN paper demonstrated that a GAN can only reach a single Nash equilibrium: that's when the generator produces perfectly realistic images, and the discriminator is forced to guess (50% real, 50% fake). This fact is very encouraging: it would seem that you just need to train the GAN for long enough, and it will eventually reach this equilibrium, giving you a perfect generator. Unfortunately, it's not that simple: nothing guarantees that the equilibrium will ever be reached.
- The biggest difficulty is called **mode collapse**. Mode collapse occurs when the diversity of the generator's outputs diminishes. For instance, if the generator becomes adept at producing convincing images of one class, such as shoes, it may prioritize generating more of these images. Consequently, the discriminator focuses primarily on distinguishing fake images of that class, neglecting others. Eventually, the generator shifts to generating a different class, leading to a cycle of limited proficiency across multiple classes.

Vanishing gradients

- Vanishing gradients in GANs refer to a situation where the gradients propagated through the network during training become extremely small, particularly noticeable in the early stages of the training process.
- This phenomenon can affect the effective optimization of the network parameters, causing the learning process to slow down significantly or even come to a halt. Essentially, as the gradients diminish in magnitude, they fail to provide meaningful information for updating the model parameters, thereby hindering the network's ability to learn and improve over time.
- This issue is especially problematic in GANs, where the generator and discriminator rely on accurate gradient information to effectively compete and evolve towards producing high-quality generated samples.

- Addressing vanishing gradients typically involves careful initialization of network weights, the use of appropriate activation functions, and techniques such as gradient clipping or batch normalization to stabilize training and mitigate the impact of vanishing gradients.
- Moreover, because the generator and the discriminator are constantly pushing against each other, their parameters may end up oscillating and becoming unstable. Training may begin properly, then suddenly diverge for no apparent reason, due to these instabilities. And since many factors affect these complex dynamics, GANs are very sensitive to the hyper-parameters: you may have to spend a lot of effort fine-tuning them.

Ethical considerations related to GANs

- **Deceptive Uses:** GANs enable the generation of highly realistic synthetic media, raising concerns about their potential misuse for deceptive purposes such as creating fake news, misinformation, or fraudulent content.
- **Biases in Generated Data:** GANs learn from existing datasets, which may contain biases present in the real world. This can lead to the propagation or amplification of societal biases, reinforcing existing stereotypes or discrimination in generated data.
- **Privacy Concerns:** GANs can generate photorealistic images of individuals, raising privacy concerns regarding the creation of synthetic images without consent. These images could be used for various malicious purposes, including identity theft, harassment, or unauthorized surveillance.

- **Regulatory and Legal Frameworks:** The rapid advancement of GAN technology necessitates the development of robust regulatory and legal frameworks to address potential ethical challenges and ensure responsible use. This includes considerations of intellectual property rights, data privacy regulations, and accountability for generated content.
- **Mitigation Strategies:** Ethical considerations surrounding GANs underscore the importance of implementing mitigation strategies such as transparency, accountability, and ethical guidelines for researchers, developers, and users. This includes promoting awareness of the ethical implications of GAN technology and fostering interdisciplinary collaboration between technologists, ethicists, policymakers, and civil society to address emerging challenges.